Nuclear Energy in the Neighbourhood? Examining Public Support for Small Modular Nuclear Reactors in the United Kingdom

Timothy B. Gravelle[1], Stuart Bagnall[2], Antonios Katris[2], Thomas J. Scotto[2], Frankie Watson-Brown[2], and Jason Reifler[3]

[1]Wilfrid Laurier University
[2]University of Strathclyde
[3]University of Southampton

October 2026

Abstract

The United Kingdom faces growing demands on its electricity grid and electricity prices as a major cost of living issue. At the same time, the UK has made a commitment to transition away from carbon-emitting sources of energy. Nuclear energy is one proposed solution, specifically small modular reactors – a newer technology with a smaller geographic footprint than traditional nuclear energy. This paper asks a set of related questions that centre on public support for small modular nuclear reactors in the UK. Does the public support their development? To what extent does public opinion shift as a consequence of describing modular nuclear reactors as a *power station* as opposed to a *reactor*? Does presenting modular nuclear reactors as a local (versus a national) issue shift support? How does geographic proximity to existing nuclear energy sites shape support?

The paper explores these questions by drawing on original survey data collected in the UK (n = 1,276). Results indicate that the UK public is generally supportive of small modular nuclear reactors. Still, using the term *reactor* instead of *power station* causes support to decrease. Support also decreases when the issue is framed in terms of developing small modular nuclear reactors in respondents' local areas as opposed to the UK. Support is also lower in those parts of the UK near existing nuclear sites as opposed to those further away. There are also regional dynamics at work, with lower support in Scotland compared to England, especially when the question posed relates to developing modular nuclear reactors in the local area.

Keywords: Public opinion, nuclear energy, modular reactors, United Kingdom

Introduction

Energy production and energy costs have been salient political issues in recent years in the United Kingdom. Energy costs in the UK have long been among the highest in the OECD. Geopolitical crises like the Russian invasion of Ukraine in 2022 and the 2026 US-Iran conflict have only served to drive them higher, becoming a major cost of living issue. Household subsidies for energy bills have thus been a persistent and contentious political issue for successive Conservative and Labour governments in Westminster from the Energy Price Guarantee introduced in 2022 to ongoing debates about targeted assistance with energy costs versus universal supports. Indeed, the terms "energy debt" and "energy poverty" have become part of the national conversation in the UK alongside debates about energy security.

At the same time, the UK remains legally committed to net zero emissions by 2050 under the amended Climate Change Act 2008, even if this remains politically contested. To achieve net zero, renewable and low-carbon sources of energy will have to be part of the solution. This includes nuclear energy, specifically small modular nuclear reactors (SMNRs). Modular reactors are a new technology designed to be more compact and require a smaller footprint than conventional nuclear power stations, with components built offsite to allow for quicker assembly onsite. In 2025, the UK government chose Rolls Royce as the successful proponent of its Great British Nuclear (GBN) competition to design SMNRs. It also established the Nuclear Regulatory Taskforce whose mandate included the streamlining of regulatory processes relating to civil nuclear projects as well as developing plans to engage civil society in siting plans for SMNRs (Jones, Gheorghiu, & Smith, 2026; Osiolo & Mukundan, 2026).

In this context, it is worth asking: what does the UK public think, and what factors shape individual-level support for (or conversely opposition to) SMNRs? While public attitudes toward nuclear power generation is a long-standing topic of inquiry (Aina & Poy, 2026; Bickerstaff, Lorenzoni, Pidgeon, Poortinga, & Simmons, 2008; Corner et al., 2011; Venables, Pidgeon, Parkhill, Henwood, & Simmons, 2012), there are presently few studies examining attitudes toward modular reactors (but see Durdovic et al., 2025; Jones et al., 2026; Kiviluoma, Savela, & Kojo, 2025). This article explores a set of questions relating to support for SMNRs in the UK. First, what is the level of public support for SMNRs? Second, is support shaped by how the technology is framed, either as a power station or as a reactor? Third, does framing SMNRs as a local issue versus a national one shift support? Fourth, does spatial proximity to

existing nuclear sites shape support? To do so, it draws on an original survey-based experiment (Druckman, 2022; Mutz, 2011) conducted with a large sample (n = 1,276) of the UK public.

To preview the main findings, SMNRs enjoy high levels of support among the UK public. Still, framing in communication induces modest shifts in expressed levels of support: power stations receive higher support than reactors, and the development of SMNRs receives lower support when it is describe as being in one's local area compared to when it is described as being in the UK. The results also indicate varied regional dynamics: support for SMNRs is lower in Scotland compared to England, particularly when SMNRs are framed as being developed in one's local area. Further, proximity to exisiing nuclear infrastructure dampens support for SMNRs.

Public Support for Nuclear Energy: Existing Research

As a relatively mature technology, attitudes toward nuclear power generation have received less scholarly attention in recent years compared to wind, solar, fossil fuels or electricity transmission lines (Carley, Konisky, Atiq, & Land, 2020).

Theoretical Expectations

Analysts of public opinion have long recognised that attitudes toward particular objects are shaped by how they are depicted and discussed – that is, how they are framed. In general terms, frames are the ways in which communication about an object (whether a policy, a technology, a social issue, or anything else) selects or emphasises particular aspects related to that object, rendering these aspects more salient (Entman, 1993). Framing effects are shifts in individual-level attitudes that are the consequence of communications that activate (or assign greater importance to) different considerations (Chong & Druckman, 2007; Druckman, 2001; Nelson, Oxley, & Clawson, 1997). Specifically in survey settings, responses are shaped by the frames employed in the questions asked (Zaller & Feldman, 1992).

Frames are certainly relevant to energy issues. For example, Clarke and colleagues (2015) find that *fracking* is seen as having greater risks, more negative consequences, and has less public support than *shale oil and gas development*. Stokes and Warshaw (2017) find that job benefit and air pollution reduction frames increase support for renewable energy. In the case of nuclear energy, frames are equally relevant. There are long-standing frames present in the news

media. Seminal work by Gamson and Modigliani (1989) identified a set of frames employed in media coverage of nuclear power, the most prevalent being technological progress and what they termed "runaway" – the notion that nuclear energy had outrun the ability of humankind to control it, and obviously intended to invoke an uncontrolled (or runaway) nuclear chain reaction. These progress and risk frames have been studied from other angles. Using a word association task and the stimulus term "nuclear power," Slovic (2000) and colleagues find that responses having negative associations outweighed positive ones. Outside of references to nuclear warfare, the most common negative associations related to nuclear waste and nuclear accidents or meltdowns (Slovic et al., 2000). Further, Bolsen and colleagues (2014) find that a health risk frame decreases support for nuclear power. Qualitative evidence indicates that such risks are important consideration in how citizens think about nuclear energy (Bickerstaff et al., 2008).

We build on previous work by testing alternative terms: small modular nuclear *reactors* and small modular nuclear *power stations*. As with the alternative terms *fracking* and *shale oil and gas development* where *fracking* carries more negative connotations and is more closely linked environmental and health risks (Clarke et al., 2015), so too is *reactor* while *power station* is more benign. The term *reactor* evokes the risks associated with nuclear fission and uncontrolled chain reactions, whereas a *power station* is a relatively mundane feature of the built environment. Our first hypothesis is thus:

H_1: Support for small modular nuclear reactors will be lower than for small modular nuclear power stations.

Mass public opinion is often shaped by features of the built environment. Indeed, a large body of literature has examined how public support for (or opposition to) different forms of energy exploration, generation, and transmission is shaped by *proximity* to energy sites (see, e.g., Aldrich, 2013; Boudet, Zanocco, Howe, & Clarke, 2018; Clarke et al., 2016; Evensen & Stedman, 2016; Gravelle & Lachapelle, 2015; Konisky, Ansolabehere, & Carley, 2020; Latré, Perko, & Thijssen, 2017; Mueller, 2019; Mueller, Keil, & Bauer, 2017). The focus on proximity effects is well motivated: individuals more readily draw on cues from what is *near* in forming their opinions and attitudes than what is *far away*. Psychological research on construal level theory (Fujita, Henderson, Eng, Trope, & Liberman, 2006; Henderson, Fujita, Trope, & Liberman, 2006; Trope & Liberman, 2010; Trope, Liberman, & Wakslak, 2007) posits that the mental

representations of events and objects that are proximate (in either space or time) are concrete and detailed; those that are distant are abstract.

Still, existing research is equivocal on the question of whether and how proximity matters to mass public attitudes toward energy issues. On the one hand, proximity may act to heighten perceptions of the risks associated with various technologies (Boudet et al., 2018; Venables et al., 2012). On the other hand, proximity may serve to highlight localised benefits, such as jobs tied to energy generation or the construction of energy infrastructure (Aldrich, 2013; Gravelle & Lachapelle, 2015). For example, Gravelle and Lachapelle (2015) find that proximity to the proposed route of the Keystone XL pipeline increases support, principally among self-described liberals (while conservatives are consistently in favour whether near or far from the pipeline); this finding is largely confirmed by Clarke and colleagues (2016). Conversely, Mueller and colleagues (2017) and Mueller (2019) find that proximity to electricity grid projects reduces support for grid expansion. Alcorn and colleagues (2017) find no consistent effect on proximity to fracking sites on support, a finding in line with Konisky and colleagues (2020) who examine a range of energy generation and transmission projects.

Studies of attitudes toward nuclear energy exhibit the same sort of disparate findings as other sources of energy. While Greenberg (2009), McBeth and colleagues (2023) and van der Pligt and colleagues (1986) find that residents near a nuclear site had more positive attitudes toward nuclear energy than those further away, a cross-national study by Latré and colleagues (2017) finds that the Fukushima nuclear accident induced a negative shift in public attitudes toward nuclear energy, with larger negative shifts in countries closer to Japan. By contrast, Aina and colleagues (2026), Cale and Kromer (2015), and Konisky and colleagues (2020) find no effect of proximity to nuclear sites on attitudes toward nuclear energy. Two factors may underlie such varied results. The first is that proximity/distance is operationalised by different authors in different ways. In many studies, "near" is operationalised as residents within a given radius of a nuclear site (ranging from 1 to 50 miles); few studies use actual geodetic distances. Second, the localised risks and economic benefits of particular nuclear sites may vary across local contexts, thus shaping the balance of local considerations relating to opposition or support.

Most existing research on proximity effects and support for nuclear energy relate to existing nuclear sites. Neither nuclear sites nor where one lives is entirely random, however. An alternative research strategy is to experimentally manipulate the salience of nuclear energy in one's milieu. Experimental research that has done so has produced results that stand in

contrast to the equivocal character of findings of observational survey research. In survey-based experiments, Biten Butorac and colleagues (2026) and Seo and Rhee (2026) both find that describing a proposed nuclear plant as nearby as opposed to further away reduces support for its construction. Thus, for citizens, most of whom have little previous, direct contact with nuclear energy, and for whom it is literally a distant consideration, are less favourable toward building new nuclear energy infrastructure nearby. These results are further supported by observational research showing greater support for modular nuclear energy at a more distant site when respondents are also asked about support for construction nearby (Durdovic et al., 2025; Hlavacek, Cabelkova, Pawlak, & Smutka, 2023; Jones et al., 2026; Kiviluoma et al., 2025).

Considering the sum total of existing research findings, we follow recent research and advance a directional hypothesis indicating that mass publics exhibit less support for modular nuclear energy when it is described as being nearby (as opposed to further away).

H_2: Support for modular nuclear energy will be lower when it described as being in one's local area compared to being in the United Kingdom.

At same time, we consider it appropriate to advance a non-directional (two-tailed) hypothesis relating to the association between (actual) distance to the nearest nuclear site and support for modular nuclear energy.

H_3: Spatial distance to a nuclear site will be associated with support for modular nuclear energy.

Data and Methods

To test these theoretical expectations, we undertake a survey-based experiment (Druckman, 2022; Mutz, 2011). Our experiment was administered to a sample of 1,276 adults residing in the United Kingdom. Participants were members of the Vox Pop Labs online panel. As a civic engagement, digital technology, and research firm, Vox Pop Labs recruits into its online panels principally from users of its voting advice application, *Vote Compass*, who volunteer to participate in future research studies. Survey data collection took place from 7 October to 11 November 2025. The study received ethics approval from the University Ethics Committee at the University of Strathclyde.

Though our online panel participants comprise a non-probability sample, we assert that such a sample is fit for purpose given the goals of our research (cf. Baker et al., 2013). One

of our principal aims is to assess the effects of our experimental manipulations, and several studies have concluded that online non-probability samples yield estimates that do not differ substantively from probability samples (Coppock, Leeper, & Mullinix, 2018; Krupnikov & Levine, 2014; Mullinix, Leeper, Druckman, & Freese, 2015).

We nevertheless approach the task of drawing inferences from non-probability survey samples as one needing careful consideration. We thus draw on model-based approaches to weighting non-probability samples that have been advanced in the survey methodology literature. Specifically, we use the quasi-randomization method to estimate pseudo-inclusion probabilities for observations from a non-probability sample when a probability-based reference survey is available (Elliott, 2009; Elliott & Valliant, 2017; Lee & Valliant, 2009). In brief, probabilities of selection for observations from a non-probability sample can be estimated via logistic regression by pooling the data with an external probability-based survey (termed a reference survey) that contains the same set of covariates as the non-probability sample. We use the 2022–2024 Annual Population Survey (conducted by the Office of National Statistics) as the probability-based reference survey for this purpose. The initial weights obtained by taking the inverse of these estimated probabilities of selection for our Vox Pop Labs panel members are further adjusted (raked) to align the sample with the demographic and political profile of the United Kingdom general public. Each experimental condition (discussed below) is weighted separately on sex, age, educational attainment, country of birth (the UK or abroad), region, and recalled vote from the 2024 general election. This set of weighting procedures allows us to make estimates of treatment effects among the UK public.

The key elements of our survey design are as follows (we reproduce the full survey instrument in the online replication materials). The survey began with introductory questions about their attention to government and public affairs, their opinion about the most important issue facing the UK, and their knowledge and attitudes about electricity issues in the UK. Building on existing research showing that climate attitudes, trust in government, and trust in electricity companies all shape attitudes toward energy infrastructure (Aina & Poy, 2026; Konisky et al., 2020), the survey then asked participants a series of questions on these topics.

Participants were then presented with a brief description of modular nuclear reactors, and then asked a question on their level of support for them that incorporated our two experimental manipulations. These manipulations were randomized independently – i.e., in a 2×2 factorial design. Based on our expectations relating to the effect of "power station" versus "reactor"

language, the first experimental manipulation randomised the use of the terms small modular nuclear reactors (SMNRs) and small modular nuclear power stations (SMNPs). Motivated by our NIMBY expectations, the second experimental manipulation randomised the reference to modular nuclear power as either in "the United Kingdom" or "your local area." The overall design of the experimental stimulus is presented below in Figure 1, with the bold text inside square brackets to highlight which aspects of the stimulus were randomised. Following this experimental manipulation, participants were asked about their support for nuclear energy generation on a five-point response scale from strongly support (scored +0.5) to strongly oppose (scored -0.5).

[Small Modular Nuclear Reactors (SMNRs)] [Small Modular Nuclear Power Stations (SMNPs)] are a new type of nuclear **[reactor] [power station]** that can be built in factories and assembled on site. They take up less space than traditional nuclear **[reactors] [power stations]**, and are designed to provide low-carbon electricity.

To what extent do you support or oppose the development of **[SMNRs] [SMNPs]** in **[the United Kingdom] [your local area]**?

Strongly support
Somewhat support
Neither support nor oppose
Somewhat oppose
Strongly oppose
Not sure

Figure 1: Experimental stimulus

After the experimentally manipulated question on SMNRs, participants were asked questions about political party identification, voting intentions in a future general election, recalled vote from the 2024 general election, left-right ideological orientations, as well as their personal demographic characteristics.

Testing our theoretical expectation about proximity to existing nuclear energy infrastructure requires us to place our participants on the map, and then calculate their proximity to – or conversely, distance from – existing nuclear energy sites, both active and decommissioned. To do so, we assign National Grid coordinates to the survey participants using their self-reported postcodes in conjunction with data products from the Office for National Statistics (the National Statistics Postcode Lookup dataset) and the Ordnance Survey (the Code-Point with Polygons shapefile). For nuclear energy infrastructure, we draw on the site location data made available as part of the Digest of UK Energy Statistics (DUKES) annual reports published by the UK

Department for Energy Security and Net Zero for 2001 to 2024 (`https://www.gov.uk/government/collections/digest-of-uk-energy-statistics-dukes`). This allows us to collate a list of nuclear energy sites. Performing a spatial join (point to point) between these two datasets provides us with the distance (in kilometres) to the nearest nuclear site for each participant. The resulting distance values range between 1.9 and 523.0 kilometres with a mean value of 83.2 kilometres.

To evaluate our theoretical expectations, we estimate a set of linear regression models in a Bayesian statistical framework (Gill & Bao, 2024; Kaplan, 2023) with 5 chains and 10,000 post-warmup iterations for each chain using the *brms* package (Bürkner, 2017) version 2.23.0 in R version 4.6.0 as a front-end to estimation in *Stan*. Our focus is on the main effects of our two experimental manipulations along with the effect of proximity to nuclear energy infrastructure. We report below the posterior means for our coefficients (β), the posterior standard deviations (SD) as measures of uncertainty, as well as the equal-tailed 95% credible intervals (CrI) from these Bayesian models. We also report the posterior probabilities ($P(\beta)$) for specific effects where these are relevant.

Our models include a set of demographic covariates: country (that is, England, Wales, Scotland, and Northern Ireland), sex, age, and educational attainment. We also follow existing research on attitudes toward energy generation and transportation infrastructure (Aina & Poy, 2026; Konisky et al., 2020) and include model specifications that include a set of attitudinal covariates. These are: left-right ideological orientation, following news about electricity issues (as a measure of issue awareness), beliefs about the need for the UK to produce more electricity, the need for the UK to become self-sufficient in electricity, the need to keep the price of electricity in the UK low, and composite measures (created from a confirmatory factor analysis model) of climate concern, trust in government, and trust in electricity companies. We include these covariates for their domain relevance, and to improve the precision of the estimates of experimental effects (Gerber & Green, 2012, pp. 96–105; Lin, 2013).

Results

We find that public support for modular nuclear energy generation in the UK is generally favourable. Across our experimental conditions, and without adjusting for covariates, mean support ranges between 0.194 and 0.285 on our five-category (-0.5 to $+0.5$) response scale.

Turning to the effects of our experimental manipulations, the results from model 1 indicate that asking about support for modular nuclear *reactors* (as opposed to modular nuclear *power stations*) yields slightly lower support ($\beta = -0.022$, $SD = 0.014$, CrI = $[-0.050, 0.005]$, $P(\beta < 0) = 0.942$). Models 2 and 3 yield very similar results (see Figure 2). In substantive terms, public support for modular nuclear energy generation remains favourable whether the technology is described as power stations or reactors, with only a slight negative shift in the case of the latter term. This is further shown in predicted scores for the constituent parts of the UK in Figure 3.[1] In England, the predicted support for power stations is 0.235, while it is 0.212 for reactors. Predicted support in Scotland is slightly lower, with 0.196 for power stations and 0.174 for reactors. Predicted support in Wales is 0.252 for power stations and 0.230 for reactors, while it is 0.267 for power stations and 0.245 for reactors in Northern Ireland.

The results from models 1–3 further indicate that asking about support for modular nuclear energy generation in one's *local area* (as opposed to in the *United Kingdom*) softens support. Model 1 estimates the main effect of the local area–UK experimental manipulation, and this shows a negative effect for the local area framing ($\beta = -0.041$, $SD = 0.014$, CrI = $[-0.069, -0.013]$). While statistically robust, and congruent with previous research (Biten Butorac et al., 2026; Seo & Rhee, 2026), the magnitude of this effect again indicates that mass opinion in the UK overall remains broadly favourable toward modular nuclear energy generation, regardless of whether it is presented as in the UK (and potentially some considerable distance away) or in one's local area.

It is important to highlight that model 1 tests only the main effect of the local area framing. Models 2 and 3 relax this assumption by interacting the local area–UK manipulation with country, thereby testing for heterogeneous treatment effects across the constituent parts of the UK. Model 2 includes interaction terms for Scotland, Wales, and Northern Ireland, contrasting the effect of the local area framing in these parts of the UK with its effect in England. The results indicate that the local area frame has a substantially larger negative effect in Scotland compared to England. While the lower-order term (indicating the effect in England) remains negative ($\beta = -0.025$, $SD = 0.016$, CrI = $[-0.056, 0.006]$, $P(\beta < 0) = 0.945$), the higher-order Scotland $\times$ local area interaction term is large and negative ($\beta = -0.157$, $SD = 0.050$, CrI = $[-0.254, -0.060]$).

[1] These predicted scores are calculated by setting all continuous covariates to their means, sex to female, and the local–UK experimental manipulation to the *local* level.

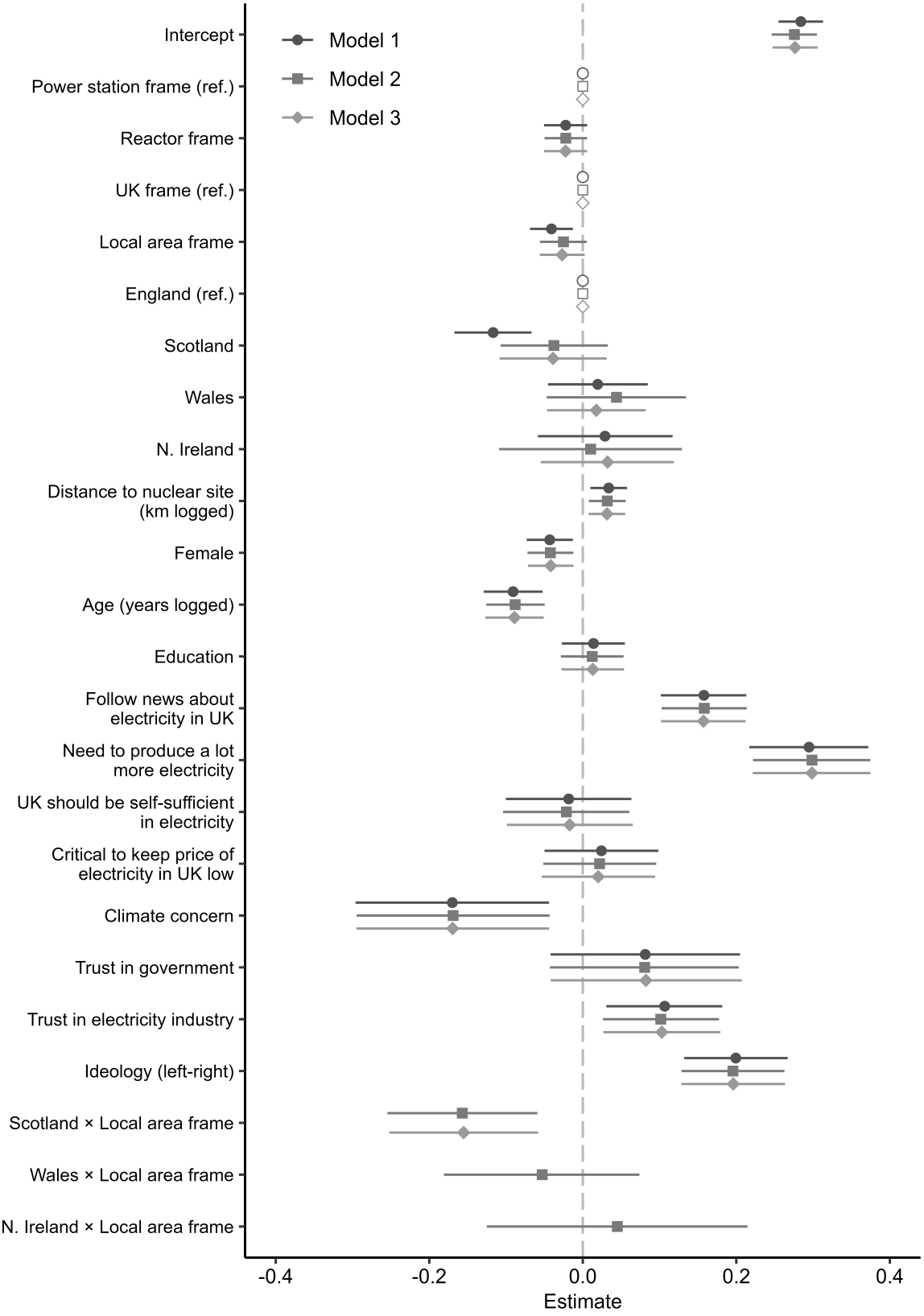


Figure 2: Coefficient plot: experimental effects and covariates

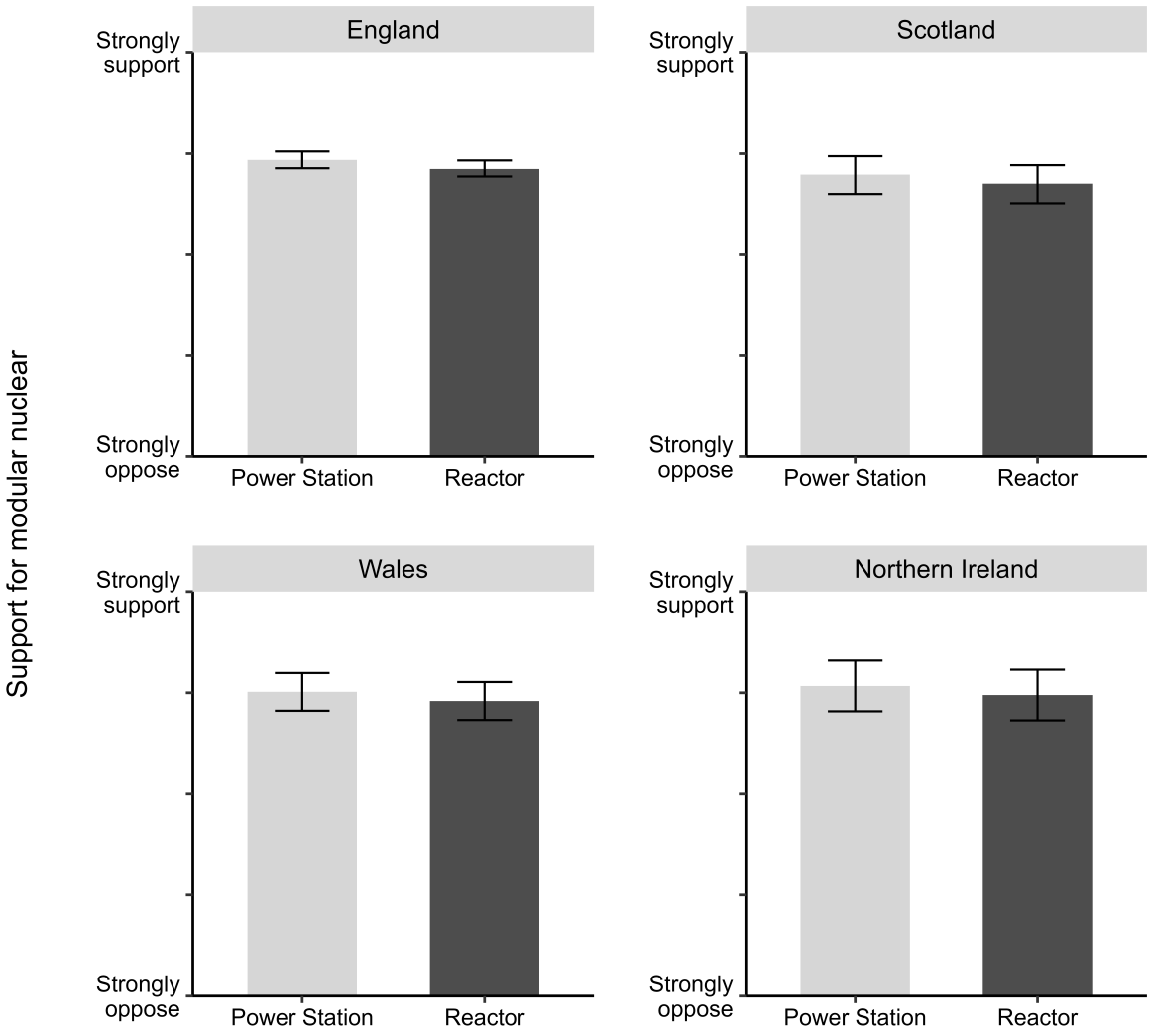


Figure 3: Predicted support for modular nuclear energy by power station–reactor framing and country

Still, the Bayes factor comparing model 2 and model 1 is only 1.014, providing little evidence to conclude that model 2 ought to be preferred over model 1.

In light of this mixed evidence on the interaction of country and the local frame from Model 2, Model 3 takes a different approach and tests a different model specification with only a Scotland $\times$ local area higher-order interaction term, imposing a common slope for the local area effect in England, Wales, and Northern Ireland. Both the small estimates and wide credibility estimates for the higher-order interaction terms for Wales and Northern Ireland – in contrast to the results for Scotland – justify doing so. This model specification yields nearly identical results: the lower order term is again negative ($\beta = -0.027$, SD = 0.015, CrI = $[-0.056, 0.002]$, $P(\beta < 0) = 0.967$) while the Scotland $\times$ local area higher-order interaction term is again strongly negative ($\beta = -0.157$, SD = 0.050, CrI = $[-0.254, -0.060]$, $P(\beta < 0) = 0.999$, $P(\text{combined } < 0) > 0.999$). The Bayes factor comparing model 3 and model 1 is 17.622, providing strong evidence that model 3 outperforms model 1. The markedly different response of Scots to the local area frame is apparent in the predicted scores shown in Figure 4.[2] The predicted support for modular nuclear energy in England is 0.235 when it is described as being in the UK and 0.208 when it is described as being in one's local area. In Wales, predicted support is at similar levels: 0.252 in the UK and 0.225 in one's local area. The same holds in Northern Ireland where predicted support for modular nuclear energy is 0.267 in the UK and 0.240 in one's local area. By contrast, predicted support in Scotland is 0.196 when it is described as being in the UK; predicted support drops to 0.014 when it is described as being in one's local area.

While the local area–UK experimental manipulation involved an attempt to consider nuclear energy generation as a proximate (as opposed to a distant) consideration, actual spatial proximity to existing nuclear energy sites also matters. In model 3, logged distance to an active or decommissioned nuclear site is positively associated with support modular nuclear energy generation ($\beta = 0.032$, SD = 0.012, CrI = $[0.008, 0.056]$); the results from models 1 and 2 are very similar. Substantively, this association indicates that support for modular nuclear energy generation increases with greater distance from existing nuclear energy sites. Conversely, closer proximity to existing nuclear energy sites is associated with reduced support. Again, generating predicted scores (in this case, focusing on England and Scotland, for participants in the *local area* and *power station* conditions) illustrates the substantive importance of proximity to nuclear sites

[2]These predicted scores are similarly calculated by setting all continuous covariates to their means, sex to female, and the power station–reactor experimental manipulation to the *power station* level.

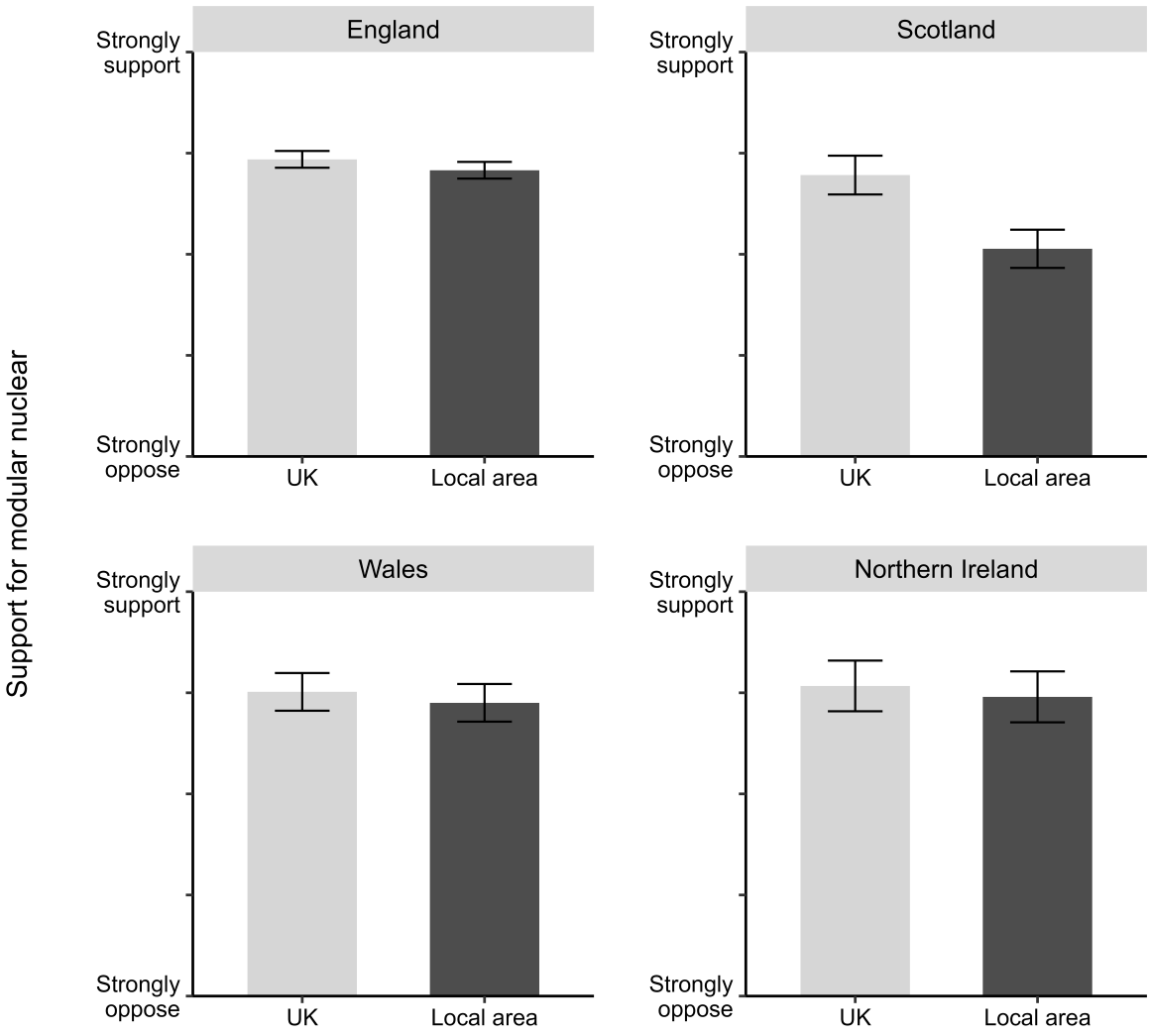


Figure 4: Predicted support for modular nuclear energy by local–UK framing and country

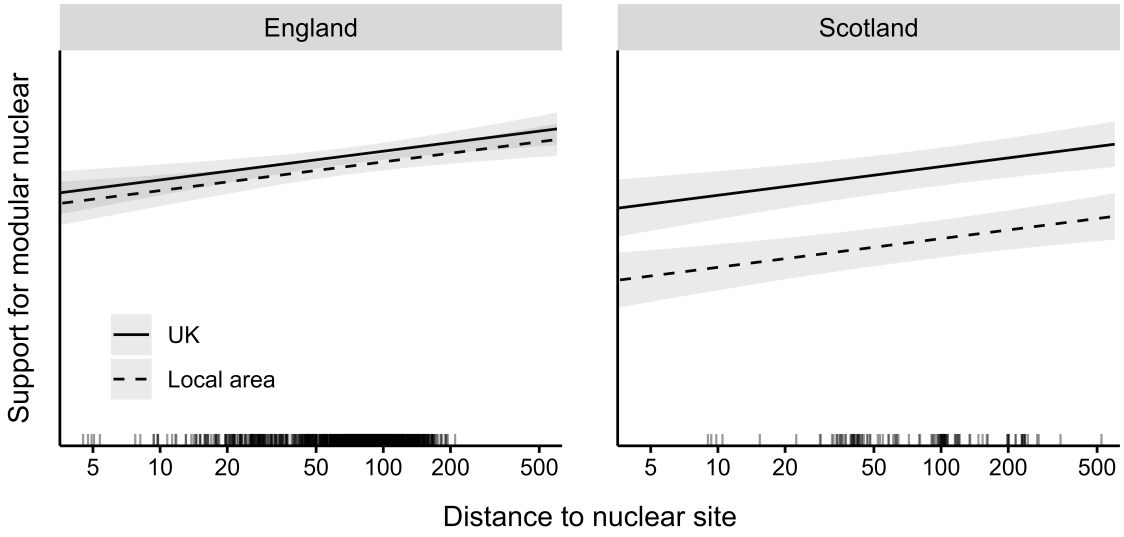


Figure 5: Predicted support for modular nuclear energy by distance to nuclear site, England and Scotland

on support (see Figure 5).[3] In England, predicted support is 0.146 at a distance of 10 kilometres, increasing to 0.196 at 50 kilometres, 0.218 at 100 kilometres and 0.240 at 200 kilometres. By contrast, predicted support in Scotland is -0.049 at a distance of 10 kilometres, 0.002 at 50 kilometres, 0.024 at 100 kilometres and 0.046 at 200 kilometres. It is worth highlighting that this association is robust in the face of several covariates, including left-right ideology, climate attitudes, trust in electricity companies, and beliefs about electricity generation.

Considering some of the observational covariates, concern over climate change is negatively associated with support for modular nuclear energy, a finding that accords with the findings of Corner and colleagues (2011) and de Groot and colleagues (2013). This maintains despite the lower CO_2 emissions associated with nuclear energy compared to fossil fuels. Also, those on the political right are more supportive of modular nuclear energy, a finding congruent with Franchino (2014), Konisky and colleagues (2020), and McBeth and colleagues (2023).

Discussion and Conclusion

Abcd.

[3]These predicted scores are again calculated by setting all continuous covariates to their means, sex to female, the power station–reactor experimental manipulation to the *power station* level, and the local–UK experimental manipulation set alternatively to the *local* and *UK* levels.

References

Aina, C., Baici, E., & Poy, S. (2026). Explaining attitudes towards nuclear energy in Italy: disentangling key drivers in a joint framework. *Energy Policy*, *216*. doi: 10.1016/j.enpol.2026.115392

Aina, C., & Poy, S. (2026). Fission and friction: A systematic review of individual-level determinants of attitudes toward nuclear energy in advanced economies. *Energy Research and Social Science*, *133*. doi: 10.1016/j.erss.2026.104587

Alcorn, J., Rupp, J., & Graham, J. D. (2017). Attitudes toward "fracking": Perceived and actual geographic proximity. *Review of Policy Research*, 1–33. doi: 10.1111/ropr.12234

Aldrich, D. P. (2013). A normal accident or a sea-change? Nuclear host communities respond to the 3/11 disaster. *Japanese Journal of Political Science*, *14*(2), 261–276.

Baker, R., Brick, J. M., Bates, N. A., Battaglia, M., Couper, M. P., Dever, J. A., ... Tourangeau, R. (2013). Summary report of the AAPOR task force on non-probability sampling. *Journal of Survey Statistics and Methodology*, *1*(2), 90–105. doi: 10.1093/jssam/smt008

Bickerstaff, K., Lorenzoni, I., Pidgeon, N. F., Poortinga, W., & Simmons, P. (2008). Reframing nuclear power in the UK energy debate: Nuclear power, climate change mitigation and radioactive waste. *Public Understanding of Science*, *17*(2), 145–168. doi: 10.1177/0963662506066719

Biten Butorac, M., Nicoli, F., & Lalli, R. (2026). Unpacking public support for nuclear energy: A twin conjoint experiment on innovative nuclear fusion and next-generation fission technologies. *Energy Policy*, *210*. doi: 10.1016/j.enpol.2025.115007

Bolsen, T., Druckman, J. N., & Cook, F. L. (2014). How frames can undermine support for scientific adaptations: Politicization and the status-quo bias. *Public Opinion Quarterly*, *78*(1), 1–26. doi: 10.1093/poq/nft044

Boudet, H. S., Zanocco, C. M., Howe, P. D., & Clarke, C. E. (2018). The effect of geographic proximity to unconventional oil and gas development on public support for hydraulic fracturing. *Risk Analysis*, *38*(9), 1871–1890. doi: 10.1111/risa.12989

Bürkner, P. C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. doi: 10.18637/jss.v080.i01

Cale, T., & Kromer, M. (2015). Does proximity matter? Plant location, public awareness, and support for nuclear energy. *Social Science Journal*, *52*(2), 1–8. doi: 10.1016/j.soscij.2015.01.002

Carley, S., Konisky, D. M., Atiq, Z., & Land, N. (2020). Energy infrastructure, NIMBYism, and public opinion: A systematic literature review of three decades of empirical survey literature. *Environmental Research Letters*, *15*(9). doi: 10.1088/1748-9326/ab875d

Chong, D., & Druckman, J. N. (2007). A theory of framing and opinion formation in competitive elite environments. *Journal of Communication*, *57*(1), 99–118. doi: 10.1111/j.1460-2466.2006.00331.x

Clarke, C. E., Budgen, D., Hart, P. S., Stedman, R. C., Jacquet, J. B., Evensen, D. T. N., & Boudet, H. S. (2016). How geographic distance and political ideology interact to influence public perception of unconventional oil/natural gas development. *Energy Policy*, *97*, 301–309. doi: 10.1016/j.enpol.2016.07.032

Clarke, C. E., Hart, P. S., Schuldt, J. P., Evensen, D. T. N., Boudet, H. S., Jacquet, J. B., & Stedman, R. C. (2015). Public opinion on energy development: The interplay of issue framing, top-of-mind associations, and political ideology. *Energy Policy*, *81*, 131–140. doi: 10.1016/j.enpol.2015.02.019

Coppock, A., Leeper, T. J., & Mullinix, K. J. (2018). Generalizability of heterogeneous treatment effect estimates across samples. *Proceedings of the National Academy of Sciences*, *115*(49), 12441–12446. doi: 10.1073/pnas.1808083115

Corner, A., Venables, D., Spence, A., Poortinga, W., Demski, C., & Pidgeon, N. (2011). Nuclear power, climate change and energy security: Exploring British public attitudes. *Energy Policy*, *39*(9), 4823–4833. doi: 10.1016/j.enpol.2011.06.037

De Groot, J. I. M., Steg, L., & Poortinga, W. (2013). Values, perceived risks and benefits, and acceptability of nuclear energy. *Risk Analysis*, *33*(2), 307–317. doi: 10.1111/j.1539-6924.2012.01845.x

Druckman, J. N. (2001). On the limits of framing effects: Who can frame? *Journal of Politics*, *63*(4), 1041–1066. doi: 10.1111/0022-3816.00100

Druckman, J. N. (2022). *Experimental thinking: A primer on social science experiments*. Cambridge University Press.

Durdovic, M., Turcanu, C., Geysmans, R., Meskens, G., Perko, T., Sala Escarrabill, R., … Brabants, W. (2025). *Public attitudes towards small modular reactors: An emerging research field and evidence from six countries.* doi: 10.20348/STOREDB/1212/1329

Elliott, M. R. (2009). Combining data from probability and non-probability samples using pseudo-weights. *Survey Practice*, *2*(6), 1–7. doi: 10.29115/sp-2009-0025

Elliott, M. R., & Valliant, R. (2017). Inference for nonprobability samples. *Statistical Science*, *32*(2), 249–264. doi: 10.1214/16-STS598

Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, *43*(4), 51–58. doi: 10.1111/j.1460-2466.1993.tb01304.x

Evensen, D., & Stedman, R. (2016). Scale matters: Variation in perceptions of shale gas development across national, state, and local levels. *Energy Research & Social Science*, *20*, 8. doi: 10.1016/j.erss.2016.06.010

Franchino, F. (2014). The social bases of nuclear energy policies in Europe: Ideology, proximity, belief updating and attitudes to risk. *European Journal of Political Research*, *53*(2), 213–233. doi: 10.1111/1475-6765.12029

Fujita, K., Henderson, M. D., Eng, J., Trope, Y., & Liberman, N. (2006). Spatial distance and mental construal of social events. *Psychological Science*, *17*(4), 278–280. doi: 10.1111/j.1467-9280.2006.01698.x

Gamson, W. A., & Modigliani, A. (1989). Media discourse and public opinion on nuclear power: A constructionist approach. *American Journal of Sociology*, *95*(1), 1–37.

Gerber, A. S., & Green, D. P. (2012). *Field experiments design, analysis, and interpretation*. W. W. Norton.

Gill, J., & Bao, L. (2024). *Bayesian social science statistics*. Cambridge University Press. doi: 10.1017/9781009341189

Gravelle, T. B., & Lachapelle, E. (2015). Politics, proximity and the pipeline: Mapping public attitudes toward Keystone XL. *Energy Policy*, *83*, 99–108. doi: 10.1016/j.enpol.2015.04.004

Greenberg, M. R. (2009). NIMBY, CLAMP, and the location of new nuclear-related facilities: U.S. national and 11 site-specific surveys. *Risk Analysis*, *29*(9), 1242–1254.

Henderson, M. D., Fujita, K., Trope, Y., & Liberman, N. (2006). Transcending the "here": The effect of spatial distance on social judgment. *Journal of Personality and Social Psychology*, *91*(5), 845–856. doi: 10.1037/0022-3514.91.5.845

Hlavacek, M., Cabelkova, I., Pawlak, K., & Smutka, L. (2023). Nuclear reactor at home? Public acceptance of small nuclear reactors in the neighborhood. *Frontiers in Energy Research*, *11*. doi: 10.3389/fenrg.2023.1211434

Jones, C. R., Gheorghiu, A., & Smith, J. T. (2026). Small reactors, big reactions: Understanding public attitudes towards small modular reactors in the UK. *Energy Policy*, *218*. doi: 10.1016/j.enpol.2026.115515

Kaplan, D. (2023). *Bayesian statistics for the social sciences* (2nd ed.). Guilford.

Kiviluoma, N., Savela, N., & Kojo, M. (2025). Exploring gender insights on the perceived safety and acceptability of small modular reactors. *Energy Strategy Reviews*, *62*. doi: 10.1016/j.esr.2025.101882

Konisky, D. M., Ansolabehere, S., & Carley, S. (2020). Proximity, nimbyism, and public support for energy infrastructure. *Public Opinion Quarterly*, *84*(2), 391–418. doi: 10.1093/poq/nfaa025

Krupnikov, Y., & Levine, A. S. (2014). Cross-sample comparisons and external validity. *Journal of Experimental Political Science*, *1*(1), 59–80. doi: 10.1017/xps.2014.7

Latré, E., Perko, T., & Thijssen, P. (2017). Public opinion change after the Fukushima nuclear accident: The role of national context revisited. *Energy Policy*, *104*, 124–133. doi: 10.1016/j.enpol.2017.01.027

Lee, S., & Valliant, R. (2009). Estimation for volunteer panel web surveys using propensity score adjustment and calibration adjustment. *Sociological Methods & Research*, *37*(3), 319–343.

Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique. *Annals of Applied Statistics*, *7*(1), 295–318.

McBeth, M. K., Warnement Wrobel, M., & van Woerden, I. (2023). Political ideology and nuclear energy: Perception, proximity, and trust. *Review of Policy Research*, *40*(1), 88–118. doi: 10.1111/ropr.12489

Mueller, C. E. (2019). Effects of spatial proximity to proposed electric power lines on residents' expectations, attitudes, and protest behavior: A replication study. *Energy Policy*, *130*, 341–346. doi: 10.1016/j.enpol.2019.04.018

Mueller, C. E., Keil, S. I., & Bauer, C. (2017). Effects of spatial proximity to proposed high-voltage transmission lines: Evidence from a natural experiment in Lower Saxony. *Energy Policy*, *111*, 137–147. doi: 10.1016/j.enpol.2017.09.023

Mullinix, K. J., Leeper, T. J., Druckman, J. N., & Freese, J. (2015). The generalizability of survey experiments. *Journal of Experimental Political Science*, *2*(2), 109–138. doi: 10.1017/XPS.2015.19

Mutz, D. C. (2011). *Population-based survey experiments*. Princeton University Press.

Nelson, T. E., Oxley, Z. M., & Clawson, R. A. (1997). Toward a psychology of framing effects. *Political Behavior*, *19*(3), 221–246. doi: 10.1023/A:1024834831093

Osiolo, H. H., & Mukundan, S. (2026). From promise to legitimacy: a survey-based study of stakeholder engagement and public acceptance of small modular reactors in the uk. *Energy Conversion and Management*, *30*, 101860. doi: 10.1016/j.ecmx.2026.101860

Seo, H., & Rhee, Y. C. (2026). When nuclear power feels close to home: Salience, risk information, and public acceptance. *Policy Studies Journal*, *54*(3). doi: 10.1111/psj.70100

Slovic, P., Flynn, J., Mertz, C. K., Poumadère, M., & Mays, C. (2000). Nuclear power and the public: A comparative study of risk perception in France and the United States. In O. Renn & B. Rohrmann (Eds.), *Cross-cultural risk perception: A survey of empirical studies* (pp. 55–102).

Stokes, L. C., & Warshaw, C. (2017). Renewable energy policy design and framing influence public support in the United States. *Nature Energy*, *2*. doi: 10.1038/nenergy.2017.107

Trope, Y., & Liberman, N. (2010). Construal-level theory of psychological distance. *Psychological Review*, *117*(2), 440–463. doi: 10.1037/a0018963

Trope, Y., Liberman, N., & Wakslak, C. J. (2007). Construal levels and psychological distance: Effects on representation, prediction, evaluation, and behavior. *Journal of Consumer Psychology*, *17*(2), 83–95. doi: 10.1016/S1057-7408(07)70013-X

van der Pligt, J., Eiser, J. R., & Spears, R. (1986). Attitudes toward nuclear energy: Familiarity and salience. , *18*(1), 75–93. doi: 10.1177/0013916586181004

Venables, D., Pidgeon, N. F., Parkhill, K. A., Henwood, K. L., & Simmons, P. (2012). Living with nuclear power: Sense of place, proximity, and risk perceptions in local host communities. *Journal of Environmental Psychology*, *32*(4), 371–383. doi: 10.1016/j.jenvp.2012.06.003

Zaller, J., & Feldman, S. (1992). A simple theory of the survey response: Answering questions versus revealing preferences. *American Journal of Political Science*, *36*(3), 579–616. doi: 10.2307/2111583

Table A1: Results: Linear Models

	Model 1			Model 2			Model 3		
	Estimate	**Est. Err.**	**95% CrI**	**Estimate**	**Est. Err.**	**95% CrI**	**Estimate**	**Est. Err.**	**95% CrI**
Intercept	0.284	(0.015)	[0.255, 0.313]	0.276	(0.015)	[0.246, 0.305]	0.276	(0.015)	[0.247, 0.306]
Country (Ref = England)									
Wales	0.020	(0.033)	[−0.046, 0.085]	0.044	(0.046)	[−0.046, 0.136]	0.017	(0.033)	[−0.048, 0.082]
Scotland	−0.117	(0.025)	[−0.166, −0.067]	−0.038	(0.036)	[−0.108, 0.032]	−0.039	(0.035)	[−0.108, 0.031]
Northern Ireland	0.029	(0.045)	[−0.059, 0.116]	0.010	(0.060)	[−0.107, 0.129]	0.032	(0.044)	[−0.055, 0.120]
Female	−0.043	(0.015)	[−0.073, −0.013]	−0.042	(0.015)	[−0.073, −0.012]	−0.042	(0.015)	[−0.072, −0.012]
Age (Years logged)	−0.091	(0.019)	[−0.129, −0.053]	−0.088	(0.020)	[−0.126, −0.050]	−0.089	(0.019)	[−0.127, −0.051]
Education	0.014	(0.021)	[−0.027, 0.055]	0.012	(0.021)	[−0.029, 0.053]	0.013	(0.021)	[−0.028, 0.054]
Power source frame (Ref. = power station)									
Reactor frame	−0.022	(0.014)	[−0.050, 0.005]	−0.022	(0.014)	[−0.050, 0.005]	−0.022	(0.014)	[−0.050, 0.006]
Location frame (Ref. = United Kingdom)									
Local frame	−0.041	(0.014)	[−0.069, −0.013]	−0.025	(0.016)	[−0.056, 0.006]	−0.027	(0.015)	[−0.056, 0.002]
Distance to nearest nuclear site (km logged)	0.034	(0.012)	[0.010, 0.057]	0.032	(0.012)	[0.008, 0.056]	0.032	(0.012)	[0.008, 0.056]
Follow news about electricity	0.158	(0.028)	[0.103, 0.213]	0.158	(0.028)	[0.102, 0.214]	0.157	(0.028)	[0.102, 0.213]
UK needs to produce a lot more electricity	0.294	(0.039)	[0.217, 0.371]	0.299	(0.039)	[0.222, 0.375]	0.298	(0.039)	[0.222, 0.375]
UK should be self-sufficient in electricity	−0.018	(0.042)	[−0.099, 0.063]	−0.021	(0.042)	[−0.104, 0.060]	−0.017	(0.042)	[−0.100, 0.065]
Critical to keep the price of electricity in the UK low	0.024	(0.038)	[−0.050, 0.099]	0.022	(0.038)	[−0.053, 0.096]	0.020	(0.037)	[−0.053, 0.094]
Trust in government (factor score)	0.081	(0.063)	[−0.041, 0.203]	0.081	(0.063)	[−0.043, 0.203]	0.082	(0.063)	[−0.041, 0.206]
Trust in electricity companies/suppliers (factor score)	0.107	(0.039)	[0.031, 0.183]	0.102	(0.038)	[0.027, 0.177]	0.103	(0.039)	[0.027, 0.179]
Concern with climate change (factor score)	−0.170	(0.064)	[−0.296, −0.045]	−0.169	(0.064)	[−0.293, −0.044]	−0.169	(0.064)	[−0.294, −0.044]
Ideology (left–right)	0.200	(0.035)	[0.132, 0.267]	0.195	(0.035)	[0.127, 0.263]	0.196	(0.034)	[0.129, 0.263]
Wales × Local frame				−0.053	(0.065)	[−0.180, 0.074]			
Scotland × Local frame				−0.157	(0.050)	[−0.254, −0.060]	−0.155	(0.049)	[−0.252, −0.059]
Northern Ireland × Local frame				0.046	(0.086)	[−0.122, 0.213]			
Observations	1,276			1,276			1,276		
Bayes R^2	0.222			0.224			0.223		
ELPD	−77.047			−76.793			−73.860		
LOOIC	154.094			153.586			147.720		
Sigma	0.252			0.251			0.251		
Bayes RMSE	0.251			0.252			0.252		

Models are linear models fit by Bayesian estimation in R (version 4.6.0) using brms (version 2.23.0) with the cmdstanr backend (Stan 2.38.0). Each model used 5 chains with 2,000 warmup and 10,000 post-warmup iterations per chain. Priors were flat priors on all regression coefficients and default weakly informative priors on the model intercept [Student-t(3, 0.2, 2.5)] and the residual standard deviation (sigma) [Student-t(3, 0, 2.5)]. Estimates are posterior means with standard deviations in parentheses (Est. Error) and 95% equal-tailed credible intervals in brackets. All Rhat values were ≤ 1.01 and all bulk and tail effective sample sizes exceeded 37,000 with no divergent transitions.